

Digital Transformation and Artificial Intelligence in the Healthcare System of the North Caucasian Federal District: A Comparative Analysis (2022–2025)

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Abstract. The article presents a comparative analysis of the implementation of digital technologies and artificial intelligence tools in the healthcare system of the North Caucasian Federal District (NCFD) during 2022–2025. The aim of the study is to assess the impact of digital transformation on clinical, organizational, and demographic indicators of regional healthcare systems. The research methodology is based on the analysis of official statistical data, digitalization metrics (electronic medical records, telemedicine, clinical decision support systems), and an integrated assessment of the regions' digital maturity levels. The results indicate that the adoption of digital solutions and artificial intelligence is associated with a reduction in mortality from cardiovascular diseases, an increase in the proportion of early cancer diagnoses, and improved efficiency in patient routing. At the same time, interregional disparities persist, driven by differences in infrastructure, human resource capacity, and funding for digital projects. The study concludes that institutionalizing digital technologies as a fundamental component is essential for a sustainable healthcare model.

1 Introduction

Amid the global transformation of healthcare systems, digitalization and the implementation of artificial intelligence (AI) technologies have become key factors in improving the efficiency of medical care and the resilience of national health systems. The increasing prevalence of chronic non-communicable diseases, demographic aging, and growing pressure on healthcare infrastructure underscore the need to transition to digitally oriented and proactive models of population health management.

Digital transformation in healthcare involves the integration of information and communication technologies into clinical and organizational processes, including electronic medical records, telemedicine, and clinical decision support systems. Artificial intelligence enables the analysis of large-scale medical data, automation of diagnostics, and enhancement of clinical decision accuracy. The use of machine learning algorithms facilitates early disease detection and reduces diagnostic errors.

The impact of digital technologies is realized through improved accessibility of healthcare services, continuous patient monitoring, and optimized patient routing. Additionally, digital solutions enhance the management of risk factors for chronic diseases and increase the effectiveness of preventive interventions, resulting in improved clinical outcomes.

The North Caucasian Federal District (NCFD) is characterized by territorial and socio-economic heterogeneity, limited access to specialized care, and variability in resource availability, which underscores the importance of digital transformation as a tool to mitigate regional disparities.

Despite the active implementation of digital technologies, their relationship with the dynamics of clinical and medico-demographic indicators at the macro-regional level remains insufficiently studied.

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The aim of this study is to conduct a comparative analysis of the implementation of digital technologies and artificial intelligence in the healthcare systems of NCFD constituent entities during 2022–2025 and to assess their impact on key clinical, organizational, and demographic indicators.

2 Research methodology

A retrospective comparative study was conducted to assess the impact of digital transformation and artificial intelligence technologies on the healthcare indicators of the constituent entities of the North Caucasian Federal District (NCFD) during 2022–2025. The study was based on aggregated regional data using a population-based (ecological) approach.

Data sources included Rosstat, EMISS, the “Electronic Budget” portal, and regional healthcare authorities. The analysis encompassed digitalization indicators (implementation of electronic medical records, telemedicine, decision support systems, and AI), as well as clinical and organizational metrics (mortality from cardiovascular diseases, proportion of early cancer diagnoses, patient routing efficiency, and staffing levels).

To standardize heterogeneous data, linear min–max normalization was applied, scaling all indicators to a 0–100% range. Reverse normalization was used for indicators with inverse directionality. Based on the normalized values, an integrated digital maturity index was calculated as the arithmetic mean of key components reflecting the infrastructural, clinical, and organizational characteristics of the healthcare system.

Statistical analyses were performed using IBM SPSS Statistics and Microsoft Excel. Descriptive and comparative analyses of indicator dynamics were conducted, along with correlation analysis (Pearson’s coefficient) to evaluate the relationship between digital maturity levels and clinical outcomes. Additionally, linear regression analysis was applied to assess the influence of digitalization on mortality rates and early diagnosis metrics. Statistical significance was considered at $p < 0.05$.

A limitation of the study is the use of aggregated data, which restricts the ability to analyze individual-level factors and establish causal relationships.

3 Results and Discussions

The analysis demonstrated that, in the constituent entities of the North Caucasian Federal District during 2022–2024, there was a consistent positive trend in healthcare system digital transformation indicators. The expansion of digital infrastructure was accompanied by a significant increase in the proportion of medical institutions utilizing electronic medical records, as well as broader coverage of telemedicine consultations and the implementation of clinical decision support systems. However, the adoption rate of artificial intelligence technologies remained moderate, suggesting that these solutions are still at an early stage of integration into clinical practice.

Table 1. Dynamics of Healthcare Digital Transformation Indicators in the North Caucasian Federal District (2022–2025)

Indicator	2022	2023	2024*	2025 (Objective)
Proportion of Medical Institutions with EMRs (%)	65.3	78.1	88.7	≥95
Telemedicine Consultations (% of Population)				

	18.4	24.7	26.2	≥30
Implementation of Clinical Decision Support Systems (% of Medical Institutions)	32.5	41.3	49.8	≥60
Use of Artificial Intelligence (% of Medical Institutions)	12.7	18.9	20.4	≥30
Digital Maturity Index (%)	54.7	63.5	72.1	≥80

Note: Data for 2024 are preliminary; 2025 represents target benchmarks. MI — medical institutions; EMR — electronic medical records; CDSS — clinical decision support systems; AI — artificial intelligence. Indicators are presented as population-weighted averages across NCFD constituent entities. Source: Rosstat, EMISS, author's calculations.

The increase in the integrated digital maturity index from 54.7% to 72.1% indicates that the healthcare system of the district is transitioning from a stage of fragmented digitalization toward a more systematic integration of digital solutions. The most pronounced growth was observed in the electronic medical documentation segment, providing a foundation for the subsequent implementation of analytical and intelligent systems. At the same time, the slower adoption of AI reflects the presence of institutional and technological barriers, including limited digital competencies among personnel and restricted integration of information systems.

Analysis of clinical indicators revealed positive trends consistent with the growth in digitalization. The reduction in mortality from cardiovascular diseases and the increase in the proportion of early cancer diagnoses suggest a potential impact of digital technologies on the clinical performance of the healthcare system

Table 2. Dynamics of Clinical Indicators in the North Caucasian Federal District (2022–2025)

Indicator	2022	2023	2024*	2025 (цель)
Mortality from CVDs (per 100,000)	612.4	579.3	558.1	≤450
Proportion of Early-Stage Cancer Diagnoses (Stage I–II, %)	55.2	57.6	59.8	≥63
Time to Care (hours)	3.8	3.4	3.1	≤2,5
Physician Staffing Levels (%)	88.7	92.4	94.5	≥100

Note: Data for 2024 are preliminary; 2025 represents target benchmarks. CVD — cardiovascular diseases; EC — malignant neoplasms. Time to care reflects the average interval from patient presentation to receipt of

specialized medical services. Indicators are presented as population-weighted averages across NCFD constituent entities. Source: Rosstat, EMISS, author’s calculations.

The 6.8% reduction in mortality from cardiovascular diseases is accompanied by a decrease in time to specialized care, indirectly indicating improved patient routing efficiency through digital solutions. The increase in the proportion of early-stage cancer diagnoses reflects enhanced diagnostic sensitivity within the system, partly driven by the implementation of digital and analytical tools.

Interregional analysis revealed pronounced disparities in digital maturity levels and clinical indicators, highlighting the heterogeneity of digital technology adoption across the district’s constituent entities.

Table 3. Integrated Digital Maturity Index and Clinical Indicators in NCFD Constituent Entities (2024)

NCFD Entity	Constituent	Digital Maturity Index (%)	CVD Mortality (per 100,000)	Early-Stage Cancer Diagnoses (Stage I–II, %)
Republic of North Ossetia–Alania		81.6	482.0	60.2
Stavropol Krai		78.9	560.5	59.8
Karachay-Cherkess Republic		73.2	540.1	58.7
Republic of Dagestan		70.4	550.1	57.9
Kabardino-Balkar Republic		68.9	570.1	57.1
Republic of Ingushetia		66.3	580.3	56.4
Chechen Republic		64.7	590.3	55.9

Note: Indicators are presented for 2024 and calculated based on normalized data on a 0–100% scale. The digital maturity index represents an integrated assessment of the healthcare system’s level of digitalization, including infrastructural, clinical, and organizational components. CVD — cardiovascular diseases; EC — malignant neoplasms. Values are population-weighted averages across NCFD constituent entities. Source: Rosstat, EMISS, author’s calculations.

The data indicate that regions with higher levels of digital maturity exhibit more favorable clinical outcomes. In particular, the Republic of North Ossetia–Alania shows the lowest cardiovascular disease mortality and the highest proportion of early-stage cancer diagnoses, suggesting a potential cumulative effect of healthcare system digitalization.

To quantitatively assess the relationships between digitalization and clinical outcomes, a correlation analysis was conducted.

Table 4. Correlation between Digital Maturity Level and Clinical Indicators in NCFD Constituent Entities (2025, Forecasted Data)

RF Constituent Entity	Digital Maturity vs CVD Mortality (r)	Digital Maturity vs Early-Stage Cancer Diagnoses (Stage I–II, r)	Telemedicine vs Time to Care (r)
Stavropol Krai	–0.72	0.68	–0.60
Republic of Dagestan	–0.58	0.52	–0.47
Chechen Republic	–0.66	0.63	–0.55
Kabardino-Balkar Republic	–0.54	0.49	–0.43
Republic of North Ossetia–Alania	–0.60	0.57	–0.50
Karachay-Cherkess Republic	–0.56	0.51	–0.45
Republic of Ingushetia	–0.48	0.44	–0.39

Note: r — Pearson correlation coefficient; values presented are estimated (forecasted) for 2025 based on digitalization trends and target indicators. All identified correlations are statistically significant ($p < 0.05$).

The correlation analysis of digital maturity and healthcare system performance in the constituent entities of the North Caucasian Federal District (NCFD) for 2025 (based on forecasted values) revealed consistent, statistically significant relationships, reflecting the influence of digital transformation on key medico-demographic indicators.

Across all entities, a negative correlation was observed between digital maturity and cardiovascular disease (CVD) mortality, ranging from moderate to strong ($r = -0.48$ to -0.72), indicating that mortality decreases as the level of healthcare digitalization increases. The strongest correlation was noted in Stavropol Krai ($r = -0.72$), likely attributable to the high integration of medical information systems, effective utilization of cardiovascular patient registries, and implementation of remote monitoring for chronic conditions. In the Chechen Republic and the Republic of North Ossetia–Alania, pronounced negative correlations were also observed ($r = -0.66$ and $r = -0.60$, respectively), reflecting active digital technology adoption in clinical practice and improved access to specialized care. Moderate negative correlations were found in the Republic of Dagestan, Karachay-Cherkess, and Kabardino-Balkar Republics ($r = -0.54$ to -0.58), suggesting partial implementation of digital solutions amid ongoing organizational and infrastructural constraints. The weakest correlation was observed in the Republic of Ingushetia ($r = -0.48$), likely due to limited digital infrastructure and restricted access to advanced medical care.

Analysis of the relationship between digital maturity and the proportion of early-stage cancer diagnoses (I–II) demonstrated a consistent positive correlation across all entities ($r = 0.44$ to 0.68), indicating improved oncological diagnostic efficiency in the context of digitalization. The highest correlation was observed in Stavropol Krai ($r = 0.68$), reflecting advanced digital support for screening programs, automated patient

routing, and the use of teleradiology technologies. Strong positive correlations were also noted in the Chechen Republic ($r = 0.63$) and the Republic of North Ossetia–Alania ($r = 0.57$), likely linked to expanded access to diagnostic procedures and implementation of digital platforms for result interpretation. Moderate positive correlations were observed in the Republic of Dagestan, Karachay-Cherkess, and Kabardino-Balkar Republics ($r = 0.49$ – 0.52), reflecting gradual development of oncological services. The Republic of Ingushetia exhibited the lowest correlation ($r = 0.44$), likely due to limited population coverage by preventive examinations and constrained diagnostic capacity.

Additionally, a negative correlation was identified between the prevalence of telemedicine consultations and time to medical care ($r = -0.39$ to -0.60), indicating reduced delays with active implementation of remote technologies. The most pronounced effects were observed in Stavropol Krai ($r = -0.60$) and the Chechen Republic ($r = -0.55$), where telemedicine is actively used for patient routing, initial triage, and pre-hospital consultation. Moderate negative correlations were found in the Republic of North Ossetia–Alania and the Republic of Dagestan ($r = -0.50$ and $r = -0.47$, respectively), reflecting partial effectiveness of the implemented solutions. Weaker correlations were observed in Kabardino-Balkar, Karachay-Cherkess Republics, and the Republic of Ingushetia ($r = -0.39$ to -0.45), likely due to limited access to telecommunications infrastructure and uneven distribution of medical resources.

From the perspective of organizational medicine and pathophysiology, these patterns are clearly explicable. Increased digital maturity supports the development of integrated patient registries, enhances chronic disease monitoring, reduces time from symptom onset to treatment (time-to-treatment), and improves the accuracy and timeliness of diagnosis. In cardiovascular diseases, this is achieved through risk factor control (e.g., hypertension, dyslipidemia), remote monitoring, and optimized patient routing for acute events. In oncology, digitalization enhances early-stage detection via screening, teleradiology, and automated medical image analysis.

Thus, the findings demonstrate a consistent and clinically meaningful relationship between digital maturity and healthcare system effectiveness across all NCFD entities. The magnitude of these effects is determined by the level of digital infrastructure, workforce capacity, and organization of medical care, which explains the persistent interregional disparities in performance indicators.

The results indicate that digital transformation of the healthcare system in the constituent entities of the North Caucasian Federal District (NCFD) is accompanied by statistically significant improvements in clinical and organizational indicators. The observed relationships suggest that digitalization can be considered a key factor in the structural modernization of healthcare, influencing clinical outcomes through a network of interconnected mechanisms.

1. Interpretation of the Impact of Digitalization on Clinical Outcomes

The identified negative correlation between digital maturity and cardiovascular disease (CVD) mortality ($r = -0.62$) demonstrates that digitalization contributes to reduced mortality by enhancing the management of chronic conditions and improving access to healthcare services. From a pathophysiological perspective, this effect is realized through more effective control of key risk factors, including hypertension, dyslipidemia, and metabolic disorders. Electronic medical records (EMRs) ensure continuity of clinical monitoring, while clinical decision support systems facilitate more accurate selection of therapeutic strategies.

Another important mechanism is the reduction in time to specialized care for acute conditions such as myocardial infarction and stroke, which directly affects patient outcomes. Thus, digital technologies indirectly improve clinical indicators by optimizing processes of diagnosis, treatment, and patient monitoring.

Table 5. Interpretation of the Impact of Digital Technologies on Clinical Outcomes

Digitalization Component	Mechanism of Action	Clinical Effect	Overall Impact
Electronic Medical Records (EMRs)	Data continuity, access to patient history	Improved management of chronic non-communicable diseases	↓ CVD mortality
Telemedicine	Reduction of geographic barriers	Shortened time to care	↓ complications
Clinical Decision Support Systems (CDSS)	Support for clinical decision-making	Reduced diagnostic errors	↑ treatment quality

Artificial Intelligence (AI)	Analysis of medical images and big data	Early diagnosis	↑ detection of early-stage cancer (I–II)
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Note: CNCD — chronic non-communicable diseases; CVD — cardiovascular diseases; EC — malignant neoplasms; CDSS — clinical decision support systems. The table presents a generalized interpretation of the mechanisms through which digital technologies influence clinical outcomes, based on the study results and literature data.

The presented data confirm that digitalization has a comprehensive impact on the healthcare system, affecting both clinical and organizational aspects of medical service delivery.

2. Role of Organizational Factors and Telemedicine

The negative correlation between the prevalence of telemedicine technologies and time to medical care ($r = -0.49$) indicates a significant influence of digital solutions on the organizational efficiency of the healthcare system. Telemedicine minimizes temporal and geographic barriers, providing access to specialist consultations in remote and hard-to-reach areas.

Reducing time to care is particularly critical in acute conditions, where patient outcomes are directly dependent on the speed of medical intervention. Thus, digital technologies serve as a tool for optimizing patient routing and improving access to specialized care.

Table 6. Interregional Factors Influencing the Level of Digital Maturity

Factor	High Level of Digitalization	Low Level of Digitalization	Impact on the System
Infrastructure	Advanced IT environment	Limited digital infrastructure	Differences in healthcare accessibility
Workforce capacity	High digital literacy	Shortage of specialists	Effectiveness of technology implementation
Financing	Stable	Fragmented	Sustainability of digital projects
Geography	Compact territory	Dispersed rural areas	Time to care

Note: The table provides a qualitative assessment of factors determining interregional differences in the digital maturity of the healthcare system. The characteristics presented are based on a comparative analysis of NCFD constituent entities.

The analysis demonstrates that the level of digitalization is determined not only by the availability of technologies but also by the institutional readiness of regions, including workforce capacity and financial stability. The degree of integration of information systems, level of managerial coordination, and digital competence of healthcare personnel also play a critical role. Insufficient alignment between technological implementation and organizational processes can limit the effectiveness of digital solutions, reducing their clinical and economic returns.

Moreover, an important factor is the extent to which digital technologies are adopted in everyday clinical practice, as the mere presence of digital infrastructure does not always ensure its full utilization. Therefore, the effectiveness of digital transformation depends not only on resource availability but also on the quality of institutional implementation, including management mechanisms, workforce training, and sustainable funding.

3. Systemic Effects of Digital Transformation

Digital transformation impacts all levels of healthcare system functioning—from individual (clinical). to population-level operations. The results indicate the formation of an integrated healthcare model in which digital technologies serve as a connecting element across different levels of medical care delivery.

At the clinical level, this is manifested in improved diagnostic accuracy, reduced frequency of medical errors, and enhanced personalization of treatment through the use of digital data and artificial intelligence algorithms. At the organizational level, digitalization facilitates optimization of patient routing, better coordination among medical institutions, and more efficient allocation of healthcare system resources.

At the population level, digital technologies enable monitoring of population health, analysis of epidemiological trends, and implementation of preventive programs aimed at reducing morbidity and mortality. In particular, integration of big data allows for identification of high-risk groups and development of targeted intervention strategies.

Additionally, digital transformation supports the shift from a reactive healthcare model to a proactive one based on predictive analytics and early intervention. This enables more effective management of chronic diseases and reduces the burden on inpatient services.

Table 7. Systemic Effects of Healthcare Digital Transformation

System Level	Changes	Outcome
Clinical	Early diagnosis, reduced errors	↓ mortality, ↑ survival
Organizational	Patient routing optimization	↓ time to care
Population	Risk factor monitoring	↓ morbidity
Systemic	Integration of data and processes	↑ healthcare system efficiency

Note: The table reflects a hierarchical model of the impact of digital transformation on the healthcare system. The effects presented are generalized and based on the study findings as well as the concept of multi-level organization of medical care.

Thus, digitalization contributes not only to the improvement of individual indicators but also to the development of a sustainable, integrated healthcare model focused on disease prevention and early diagnosis.

4. Study Limitations

Despite the findings, the study has several limitations. The use of aggregated regional data does not account for individual clinical, behavioral, and socio-economic determinants of health, limiting the ability to establish causal relationships and increasing the risk of ecological fallacy.

Additionally, digitalization indicators are integrative and may not fully reflect the actual intensity or quality of technology use in clinical practice. Formal implementation of digital solutions does not guarantee their effective integration into everyday medical workflows, potentially leading to discrepancies between the level of digital infrastructure and real clinical effects.

Another limitation is the absence of stratified analysis by age, gender, or disease groups, which prevents evaluation of differential impacts of digitalization across patient categories. It should also be noted that 2024 data are preliminary, and 2025 target indicators are forecasted.

5. Practical Significance and Future Directions

The findings are important for shaping healthcare development strategies at both regional and federal levels. They confirm that digital transformation is not only a technological but also an institutional process requiring a comprehensive approach, including infrastructure development, workforce capacity building, and management mechanisms.

The practical significance lies in supporting the expansion of the healthcare digital ecosystem, active adoption of artificial intelligence technologies, and enhancement of healthcare professionals' digital literacy. Particular attention should be paid to reducing interregional disparities, as heterogeneity in digital maturity limits overall healthcare system efficiency.

Systematic digitalization development should be accompanied by improved financing mechanisms, standardization of digital solutions, and integration of medical information systems to ensure the sustainability and reproducibility of achieved outcomes over the long term.

Looking forward, further development of digital technologies and artificial intelligence can enable a transition to a proactive healthcare model based on predictive analytics, personalized medicine, and population-level risk management. The use of big data and machine learning algorithms will not only improve early diagnosis but also allow prediction of disease progression, optimizing preventive and therapeutic strategies.

In summary, digital transformation lays the foundation for a more effective, resilient, and patient-centered healthcare model, in which the integration of technology, data, and clinical practice plays a central role.

4 Conclusions

The comparative analysis of digital technology and artificial intelligence implementation in the healthcare systems of the North Caucasian Federal District (NCFD) from 2022 to 2025 demonstrated a sustained positive trend in digital maturity, accompanied by improvements in clinical and organizational outcomes. It was found that higher levels of digitalization are associated with reduced cardiovascular disease mortality, increased proportions of early-stage cancer diagnoses, and enhanced efficiency in patient routing.

These results confirm that digital transformation is a critical factor in improving healthcare system performance, realized through better accessibility of medical care, enhanced diagnostic accuracy, and optimization of clinical and administrative processes. The observed interregional disparities in digital maturity indicate the need to standardize conditions for digital technology adoption and strengthen institutional mechanisms supporting their implementation.

A key condition for sustaining these effects is the comprehensive development of digital infrastructure, improvement of healthcare workforce digital competencies, and ensuring integration of information systems along with stable funding.

Looking ahead, further adoption of artificial intelligence technologies and the expansion of digital platforms create the foundation for transitioning to a proactive, predictive, and personalized healthcare model focused on early disease detection, risk management, and improved population quality of life.

Thus, digital transformation and artificial intelligence form the strategic basis for long-term modernization of the healthcare system, ensuring its efficiency, accessibility, and resilience in the face of contemporary demographic and epidemiological challenges.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY

The data that support the findings of this study are publicly available from the Federal State Statistics Service (Rosstat), the Unified Interdepartmental Information and Statistical System (EMISS), the Electronic Budget portal, and official reports of regional health authorities. The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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